

Wellhead Abnormal Reading

Field Manual — Setup and Troubleshooting Reference

Use case definition. A wellhead assembly controls and contains production through tubing, casing/annulus, valves, chokes, gauges, and transmitters. Abnormal behavior includes pressure or temperature outside the operating envelope, unexplained flow change, annulus communication, valve leakage, instrument disagreement, and pressure/flow mismatch. Troubleshooting distinguishes real well behavior from instrument or choke faults while protecting well integrity.

Manual setup and use. Use the approved well operating envelope, barrier policy, gauge calibration procedure, and OEM/site alarm or trip limits. Values are training/reference values.

Safety note. Do not approach a suspected leak, loosen fittings, or operate wellhead valves without authorization, permit, and approved barrier procedure.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: High wellhead pressure

Field Condition

tubing 142 barg from 95–120; casing 48 barg from 20–35; flow 1,180 m³/d from 1,600

Problem

High tubing and casing pressure indicates downstream restriction, hydrate risk, or well behavior change.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses tubing above 130 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when tubing 95–120 barg and casing 20–35 barg.

Expected Result

tubing 95–120 barg and casing 20–35 barg

Scenario 2: Low wellhead pressure

Field Condition

tubing 54 barg from 95–120; casing 12 barg; flow 720 m³/d

Problem

Low pressure with reduced flow may indicate inflow decline, valve restriction, or gauge error.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses tubing below 70 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when validated pressure and flow within approved envelope.

Expected Result

validated pressure and flow within approved envelope

Scenario 3: High temperature

Field Condition

wellhead 96 °C from 55–75; tubing 118 barg; flow 1,100 m³/d

Problem

Elevated temperature may indicate hot production, friction, or faulty sensor.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature above 85 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when validated temperature 55–75 °C or approved thermal assessment.

Expected Result

validated temperature 55–75 °C or approved thermal assessment

Scenario 4: Low temperature

Field Condition

wellhead 8 °C from 35–55; pressure 96 barg; flow 900 m³/d

Problem

Low temperature with high pressure can indicate Joule–Thomson cooling or hydrate risk.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature below 15 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when temperature above 30 °C or approved hydrate-control state.

Expected Result

temperature above 30 °C or approved hydrate-control state

Scenario 5: Annulus pressure increase

Field Condition

annulus 46 barg from 18–28; tubing 110 barg; increase 6 barg/h

Problem

Rapid annulus build-up may indicate communication or thermal expansion.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses rise above 2 barg/h.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when annulus stable within 18–28 barg after approved diagnosis.

Expected Result

annulus stable within 18–28 barg after approved diagnosis

Scenario 6: Annulus pressure decrease

Field Condition

annulus falls 28 to 9 barg in 40 min; tubing 112 barg; flow 1,450 m³/d

Problem

Unexpected annulus loss may indicate leakage or gauge failure.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses drop above 5 barg/30 min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure stable after barrier/instrument assessment.

Expected Result

pressure stable after barrier/instrument assessment

Scenario 7: Flow reduction

Field Condition

flow 680 m³/d from 1,500; tubing 128 barg; choke 55 %

Problem

Reduced flow with increased tubing pressure suggests choke/flowline restriction.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses flow below 900 m³/d.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when flow above 1,300 m³/d with approved pressure.

Expected Result

flow above 1,300 m³/d with approved pressure

Scenario 8: Flow increase

Field Condition

flow 2,250 m³/d from 1,500; tubing 82 barg; choke 55 %

Problem

Unexpected flow increase may indicate choke movement, control fault, or well change.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses increase above 20 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when flow returns to approved 1,400–1,650 m³/d.

Expected Result

flow returns to approved 1,400–1,650 m³/d

Scenario 9: Choke problem

Field Condition

command 60 %, feedback 22 %; tubing 136 barg; flow 720 m³/d

Problem

Choke position disagreement indicates actuator, stem, or feedback failure.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses feedback differs by >10 percentage points.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when feedback within 5 points of command.

Expected Result

feedback within 5 points of command

Scenario 10: Valve leakage

Field Condition

upstream 118 barg; downstream 41 barg; closed valve downstream rises 7 barg in 10 min

Problem

Pressure build-up across a closed valve indicates passing leakage.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses rise above 2 barg/10 min.

2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when downstream remains within 1 barg over 10 min.

Expected Result

downstream remains within 1 barg over 10 min

Scenario 11: Pressure fluctuation

Field Condition

tubing cycles 94–126 barg every 70 s; choke 48–58 %; flow 1,200–1,600

Problem

Cycling may be control instability, slugging, or restriction.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses swing above 15 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when swing below 5 barg.

Expected Result

swing below 5 barg

Scenario 12: Instrument disagreement

Field Condition

local tubing 112 barg; transmitter 128 barg; calibrated test gauge 114 barg

Problem

Transmitter is 14 barg high relative to calibrated reference.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses difference above 3 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when transmitter within 2 barg of reference.

Expected Result

transmitter within 2 barg of reference

Scenario 13: Frozen transmitter reading

Field Condition

pressure remains 101 barg for 6 h; local gauge 118 barg; flow changes 20 %

Problem

A static signal despite process change indicates a frozen input or impulse blockage.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses no change for >2 h with process change >10 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when signal tracks local reference within 2 barg.

Expected Result

signal tracks local reference within 2 barg

Scenario 14: Abnormal after maintenance

Field Condition

tubing 134 barg versus 108 before valve work; choke command 50 %, feedback 50 %; flow 760 m³/d

Problem

Post-maintenance lineup, trim, or impulse connection may be incorrect.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure increase above 15 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when tubing 95–120 barg and flow 1,400–1,650 m³/d.

Expected Result

tubing 95–120 barg and flow 1,400–1,650 m³/d

Scenario 15: Pressure/flow mismatch

Field Condition

tubing 86 barg; flow 1,980 m³/d from 1,450; choke 42 %

Problem

High flow at unexpectedly low pressure may indicate bad pressure or flow measurement.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses flow differs from test by >10 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when validated pressure/flow within 5 % of test.

Expected Result

validated pressure/flow within 5 % of test